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Final Report

Challenges in and Recommendations for Predicting Financial Crises in Emerging Markets

by

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I acknowledge the use of AI tools in this project. Here is a list of tools and the specific tasks for which they were used:

- **Anthropic Claude Sonnet 4.0 Thinking** (`https://www.anthropic.com/news/claude-4`) for gathering information on data sources, compiling a list of useful Yahoo Finance tickers, and suggesting relevant literature.
- **Anthropic Claude Sonnet 4.0** (`https://www.anthropic.com/news/claude-4`) for LaTeX formatting.
- **GitHub Copilot** (`https://github.com/features/copilot`) for debugging my code, refining my plots, and resolving package dependency issues.

The submitted work is my own and it reflects my own understanding and effort as I declared by signing the Academic Integrity Declaration.

Abstract

This project investigates financial crisis patterns and precursors in developed markets and applies these learnings to emerging markets, where data scarcity precludes the training of domain-specific models yet reliable predictions are required to inform policy interventions and risk management in economies shaped by external conditions. We leverage world development indicators from the World Bank, key economic indicators from the Organization for Economic Co-operation and Development and a macrohistory database compiled to study financial crises in developed markets. Crisis definitions from Laeven and Valencia and Reinhart and Rogoff are used as the ground truth. The study illustrates the data availability and quality gap that exists between developed and emerging markets, quantifies the shortcomings of model performance when trained only on developed markets and on a mix of both emerging and developed market data, and evaluates the viability of feature engineering and transfer learning to bridge this gap. Our results show that the performance of tree-based classifiers improves when trained on both developed and emerging markets but that deep learning alternatives perform best when designed to be domain-specific.

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1 Introduction

Financial crises are low frequency events with high macroeconomic and social costs that propagate globally through trade and capital-flow channels. A timely early warning system can allow policymakers to deploy macroprudential tools and fiscal buffers, central banks to calibrate interest rates, and investors to price sovereign debt and allocate capital [16]. The goal is to signal increased risk of an upcoming crisis based on variables that behaved differently prior to a crisis than in other economic periods [28].

A variety of models have been employed to predict financial crises, from threshold based indicators to binary choice logit and probit models to machine and deep learning. Early prediction of crises requires models to account for class imbalance, temporal sparsity of data, and structural changes (e.g. financial architecture reforms, cryptocurrency adoption) that can undermine the relevance of historical data. The current data landscape exhibits biases towards developed, less crisis-prone markets [19]. Figure 1 shows the country-year crisis labels since 1800 for 17 well-studied developed economies and select emerging markets in Africa, Latin America, and Asia. Crisis labels are positive two years prior to a crisis in the studied databases and negative otherwise.

Developed markets show patterns of crisis concurrence, often due to price shocks, credit crises, or asset bubbles introducing stress to the global financial market [1]. Emerging markets have less synchronized crisis patterns but also experience crises triggered by external shocks [27]. Crisis definitions are often based on data collected for developed markets. This limits the generalizability of crisis prediction methodologies to emerging markets, where data is incomplete, crisis frequencies and durations are understudied, and unique challenges such as commodity dependence, informal economies, and foreign exchange mismatches can add high uncertainty to model predictions [27].

This study explores the extent to which a model can learn useful representations of crisis precursors from the developed markets and apply them to unseen emerging market contexts, either by augmenting the training dataset with emerging market data or through transfer learning. Our results show that tree-based classifiers can achieve AUC-ROC of 0.72 and 0.70 when trained and evaluated on developed markets and emerging markets respectively. Performance improves for both test groups when models are trained on all markets but this does not hold true for deep learning alternatives.

A bi-directional Long-Short Term Memory (LSTM)-based sequence classifier, tuned on precision-based focal loss and trained and tested on developed markets can achieve AUC-ROC of 0.77 with an out-of-the-box AUC-ROC of 0.59 on emerging markets that degrades with transfer learning. This points to the dependence of predictions on factors that may not be relevant, or even tracked, in emerging markets. Models trained on developed markets may predict concurrent or lagged crises in emerging markets but struggle to learn transferable crisis precursors.

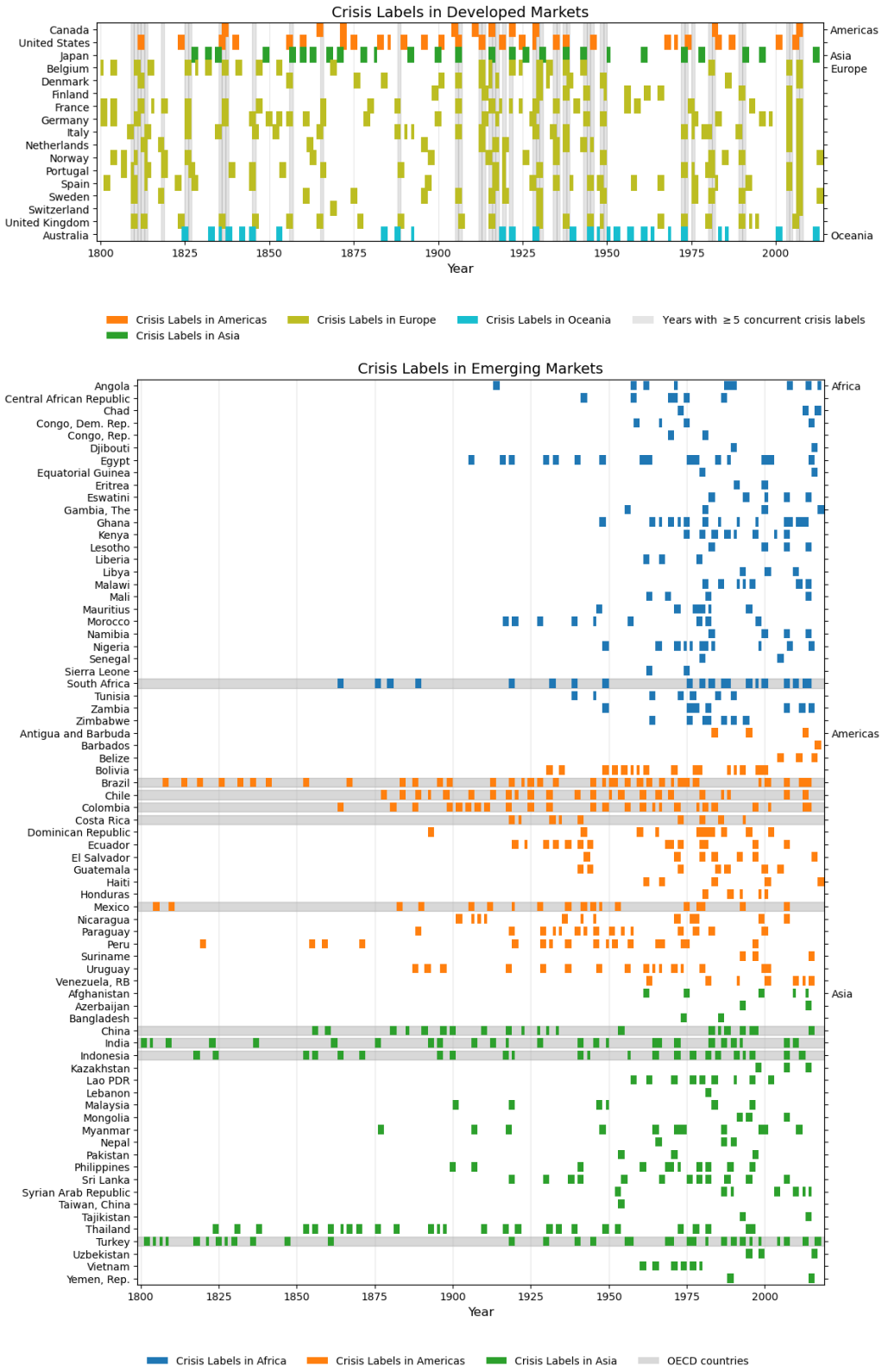

Figure 1: *Top*: each row represents one of the 17 most studied developed markets. Each colored bar is the positive crisis label, the two years prior to a financial crisis. Vertical grey stripes highlight years during which at least five developed economies have a concurrent positive crisis label. *Bottom*: each row represents one of the emerging markets that were evaluated in this study. Horizontal grey stripes highlight OECD countries. These 73 emerging markets have experienced at least two financial crises per the databases consulted.

2 Literature Review

An actionable financial crisis prediction model is one that strikes the appropriate balance between false positive and false negatives for the end user. The cost of a false negative can be high for developed markets but a false positive for emerging markets can invite speculation and unnecessary economic disruptions [15]. Here we summarize the models typically used and ways in which crisis labels, the prediction objective, can be defined.

2.1 Models

A threshold-based model flags a crisis when an economic indicator or group of indicators exceed a pre-defined level [19]. Cumulative threshold exceedances or composite scores, weighted by the inverse of the historical noise-to-signal ratio, tend to rely on theory as a guide for selection of economic features and their thresholds [20], which themselves are primarily defined by studying developed markets [21]. Binary choice models are the standard for comparison as they are explainable in log-odds space and estimate probability of crisis occurrence as a multivariate function of features [10]. They are sensitive to precision-recall thresholds and perform poorly on out-of-sample test cases [11]. Generalizability is also limited without features quantifying cross–country contagion [9].

Supervised machine learning, often in the form of decision trees, can capture more complex relationships between variables and the prediction. Ensemble methods prevent small changes in the training data from causing large changes in the classification rules [7] with Random Forests identified as strong and stable performers [18]. However, performance on test cases varies depending on the time horizon of the evaluation, potentially due to overfitting on historical trends and countries in the training data [4].

Deep learning methods such as Convolutional Neural Networks, Recurrent Neural Networks (particularly LSTM), and Transformers sometimes exhibit better performance in high dimensional feature spaces and can account for temporal contexts and dependencies [34]. Recurrent gates allow models to assess lags and interaction terms dynamically instead of relying on engineered features [34]. Autoencoders, when trained on tranquil economic periods, can detect anomalous potentially pre-crisis periods by flagging high reconstruction error [3]. These deep learning models outperform baseline logit models on developed market data [34] but require large amounts of high-frequency labeled training data which is limited to a few well-studied developed markets and can be sensitive to data preprocessing and hyperparameter choices [5]. This highlights the difficulty in the prediction problem: pre-crisis periods often do not consistently differ from normal economic periods [14], especially when relating developed to emerging markets.

2.2 Crisis Labels

Global financial crisis labelling is performed by economists and policymakers as they reflect on the economic states of countries at different points in history. There is strong agreement between studies on when developed market crises began but emerging markets are more contentious [6], [35], [29]. Oftentimes emerging market crises occur after developed or other emerging market shocks with variable delay depending on trade and debt exposure [23].

As the goal is to predict an upcoming crisis, crisis labels are usually positive before the crisis, with lead times varying depending on the market and end user. Binary classification can be evaluated with the cost of false positives and false negatives to serve the risk-appetite and aversion of the end user. Multiclass classification models distinguish between different levels of crisis severity. One approach to divide the positive label into severe and mild is by using narrative information from past crises. This is subjective, unscalable, and can introduce categorization bias [32]. A data-driven approach can use historical exceedances or rates-of-change in economic outcomes but requires sufficient data at an

appropriately high resolution and may struggle to create groups that are historically and geographically comparable, especially following financial structure reforms [25].

In sum, the use of deep learning models for emerging market crisis prediction is understudied, owing to sparse data and high asynchronicity of crisis events. Furthermore, the degree to which models trained on developed markets can learn domain-agnostic crisis precursors or adapt to emerging market contexts remains an open question.

3 Methodology

3.1 Data Sources

A labelling system may inflate or obscure a causal or coincidental link between a crisis event and reported economic variables based on the thresholds used to define crises [22]. To avoid introducing such bias into this study, ground truth labels for financial crises were collected from two databases but the features for prediction were compiled from sources that were not consulted in the creation of the labels. The crises labels were defined mainly using bank system, loan and debt default, and qualitative indicators, whereas the features for prediction focused on macroeconomic health.

The following databases were used as ground truth because they are reputed and routinely extended.

1. Laeven and Valencia [22]: A database of banking, currency, and sovereign debt crises reported for developed and emerging markets. This was extended by Nguyen et al. 2020 [25] up to 2019. Banking crises are periods showing significant signs of distress in the banking system (bank runs, losses, and/or liquidations) and significant policy interventions. Currency crises occur when nominal depreciation of a currency against USD is at least 30% a year and higher than the previous year's depreciation by at least 10% based on either average or end-of-period exchange rates. Sovereign debt crises occur when a country either fails to pay its interest by the due date or postpones its obligations. Defaults must exceed 7% of GDP or 1% of GDP over three consecutive years. Per these definitions, there have been 151 banking, 414 currency, and 200 sovereign debt crises from 1970 to 2019.

2. Reinhart and Rogoff [30]: A database of financial crises for developed markets since 1800 based on narrative accounts by economic historians. Banking crises are marked by the closure, merging, takeover, or large-scale government assistance of an important financial institution, which may or may not be spurred by a bank run. This offers a contrasting perspective to the policy-based definition above, which can be preordained, subjective, or untimely [2]. This database is updated and maintained by Harvard Business School [15].

Per an analysis of leading and coincidental indicators of financial crises [31], there are ten groups of crisis predictors: measures of output and capacity utilization, consumption and sales inventories and orders, interest rates and asset prices, exchange rates and foreign trade, employment, earnings and measured of the labor force, wages and prices, measures of government fiscal activity, and other business condition indicators. The following open source datasets were merged to develop features for prediction. These were chosen as they are cross-country, reputable, complementary, and provide features that span all ten groups.

- World Bank (WB) - World Development Indicators [33]: A global collection of 1,600 indicators from officially recognized international sources. Provides high-quality, internationally comparable statistics related to the local economy, governance, and global economic linkages, covering 217 economies since 1960. Data from Economy & Growth, Financial Sector, and External Debt topics were compiled.

- Organization for Economic Co-operation and Development (OECD) - Key Economic Indicators [26]: Monthly and quarterly data providing comparable short-term economic statistics across OECD member and select non-member countries. Covers variables including national accounts, business surveys, consumer opinions, retail sales, production, consumer prices, employment, wages, international trade, and balance of payments, which are either unavailable in WD or measured at lower resolution.

- Jordà-Schularick-Taylor (JST) Macrohistory Database [24]: A macrofinancial data collection effort focused on 17 developed economies. Unique variables include bank credit to the non-

financial private sector, mortgage lending and long-term returns on housing, equities, bonds and bills.

3.2 Experiment Design

Data preparation for this project followed the steps outlined by Bluwstein et al. 2020 [5]. As the aim was to predict a crisis ahead of time, the crisis label was positive two years before the beginning of the event and negative otherwise.

Table 1 shows that the compiled suite of features exhibited low completeness of information, particularly for emerging markets. To evaluate how variable selection and imputation affected downstream predictive performance, two data pipelines were created, producing data at the annual and monthly frequency respectively.

Table 1: Data Quality and Completeness Overview

Source	Country Group	Countries	Variables	Dates, Frequency	Completeness
WB	Developed	17	455+	1960–2025, Annual	51.7%
	Emerging (Africa)	33	511+	1960–2025, Annual	35.1%
	Emerging (Americas)	22	511+	1960–2025, Annual	47.5%
	Emerging (Asia)	29	511+	1960–2025, Annual	33.1%
OECD	Developed	17	37	1914–2024, Monthly	32.5%
	Emerging (Africa)	1	61	1957–2024, Monthly	32.1%
	Emerging (Americas)	5	61	1957–2024, Monthly	25.7%
	Emerging (Asia)	3	61	1951–2024, Monthly	25.4%
JST	Developed	17	55	1870–2020, Annual	78.2%
	Emerging (Africa)	0	—	—	—
	Emerging (Americas)	0	—	—	—
	Emerging (Asia)	0	—	—	—

3.2.1 Annual Resolution

Baseline experiments used data aggregated at country-year level. Data from WB and JST were merged. The WB dataset was the most comprehensive in its coverage of countries and variety of features. As it is a reputed source, all values were assumed to be accurately reported. Features and years with low completeness were dropped. Remaining missing values were filled using history-based k-Nearest Neighbors (KNN). This was performed for country and feature combinations where missing data was interspersed between valid observations and required interpolation without look-ahead bias. Each row was matched with its k-nearest historical neighbors which could be any point in the country's history or any country at any time in history up to and including the date with missing data. The missing value was then imputed with the inverse average of the nearest neighbors by Manhattan distance.

For developed markets with rich data but with a reporting lag, the latest values were imputed using ARIMA. For emerging markets with limited historical precedent to train ARIMA models, history-based KNN was used with larger k that allowed developed markets to factor into KNN. Any features that were unreported for any country were filled with zero. For emerging markets, this included all the variables in the JST database. The dataset was then trimmed to 1962 to 2024, with 227 features for 17 developed markets and 73 emerging markets.

3.2.2 Monthly Resolution

Higher resolution economic signals were used to test both time-invariant and time-varying models. Data prepared at annual resolution was forward filled 11 months. Monthly data was sourced from OECD to add features for OECD-members. Gaps were filled with history-based KNN. For dates with insufficient historical precedent to perform history-based KNN, values were imputed with the global median for that month or zero. Data for non-OECD emerging markets was filled with zero.

Additional feature engineering was performed on the monthly resolution OECD columns. This included creating rolling statistics (mean, standard deviation, minimum, maximum), exponentially weighted averages, distance from the historical mean, a boolean for exceeding historical extremes, momentum, volatility, and seasonal additive decomposition into trend, seasonal, and residual components.

The final dataset comprised 400 features for 17 developed markets and 73 emerging markets, with monthly observations from 1960 to 2024 for developed markets and 1962 to 2024 for emerging markets.

3.2.3 Train-Test Split

Datasets were used to train time-invariant and time-varying crisis prediction models. A train-test split was performed on both with a cutoff date of 2007-01-01. This was to include the Global Financial Crisis as a test benchmark, ensuring that the test set contained sufficient crises to evaluate model performance.

Time-invariant models took a snapshot of economic information at country-year or country-month level and predicted whether the crisis label was positive. Following the methodology in Bluwstein et al. 2020 [5] and similar studies, data from in-crisis years were excluded from the train set to avoid post-crisis bias and to ensure that a negative label was not associated with both healthy and in-crisis or recovering periods. It is noted that crises in developed and emerging markets have different patterns of duration, frequency, and recovery [27]. After initial exploration, it was decided to exclude up to four years following a positive label or until the next positive label. This ensured that crises in quick succession were individual events to be predicted. This was also necessary to enrich the minority class representation, particularly for emerging markets.

Table 2: Train–Test Split Summary

Resolution, Model	Dataset	Split	Shape	Positive Labels
Annual, Time-invariant	Developed	Train	(622, 227)	19.94%
		Test	(289, 227)	6.92%
	Emerging	Train	(2618, 227)	20.44%
		Test	(1241, 227)	9.91%
Monthly, Time-invariant	Developed	Train	(7461, 400)	20.75%
		Test	(3468, 400)	6.92%
	Emerging	Train	(28798, 400)	21.87%
		Test	(14892, 400)	9.91%
Monthly, Time-varying	Developed	Train	(9588, 400)	16.15%
		Test	(3468, 400)	6.92%
	Emerging	Train	(38617, 400)	16.31%
		Test	(14892, 400)	9.91%

Time-varying models received a sequence of consecutive monthly data and predicted whether the crisis label was positive in the final month of the sequence. Crisis periods were not removed for the

time-varying case so that the models received information on crisis duration and frequency embedded within the sequence.

Table 2 summarizes the data dimensions and crisis label distributions following the split. Figure 2 shows examples of raw and engineered training data variables.

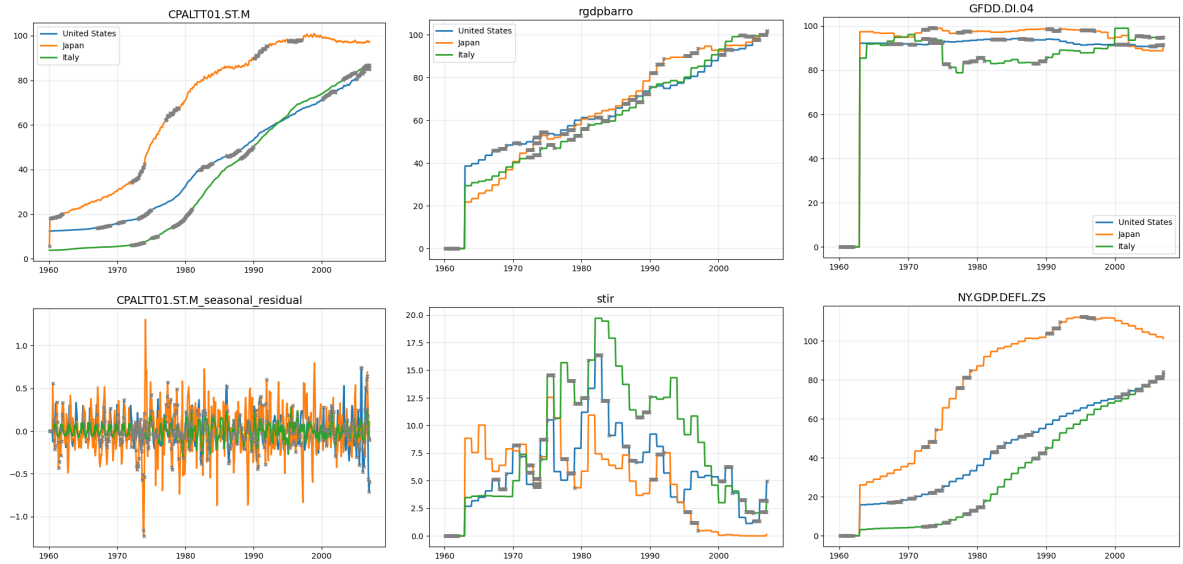

Figure 2: Example training data at monthly resolution with positive crisis labels marked. *Column 1*: Example of an OECD variable and a feature engineered from it. *Column 2*: Examples of two JST variables. *Column 3*: Examples of two WB variables.

3.2.4 Model Types

Experiments were designed to evaluate time-invariant and time-varying models on annual and monthly resolution datasets. We trained each data resolution and model combination on emerging and developed markets separately and together. Time-invariant models evaluate how distinct pre-crisis periods are from other economic states. Time-varying models evaluate how distinct pre-crisis trends are from other economic trends. Training on a combination of developed and emerging markets tests if diversity in training samples improves performance on out-of-sample tests or if it compromises the models ability to fit to domain-specific data. AUC-ROC and the quality of actual predictions were used to evaluate performance in the test period.

4 Results

4.1 Time-invariant Models

4.1.1 Machine Learning Classifiers

Random Forest and Gradient Boosting models were tested for time-invariant crisis label prediction, each taking a snapshot of economic information at country-year or country-month level and predicting whether the crisis label is positive. Figure 3 shows that AUC-ROC of the best performing model in each group is consistently greater for developed markets, which have higher completeness of data. AUC-ROC is also greater at monthly resolution, where OECD variables add high predictive value.

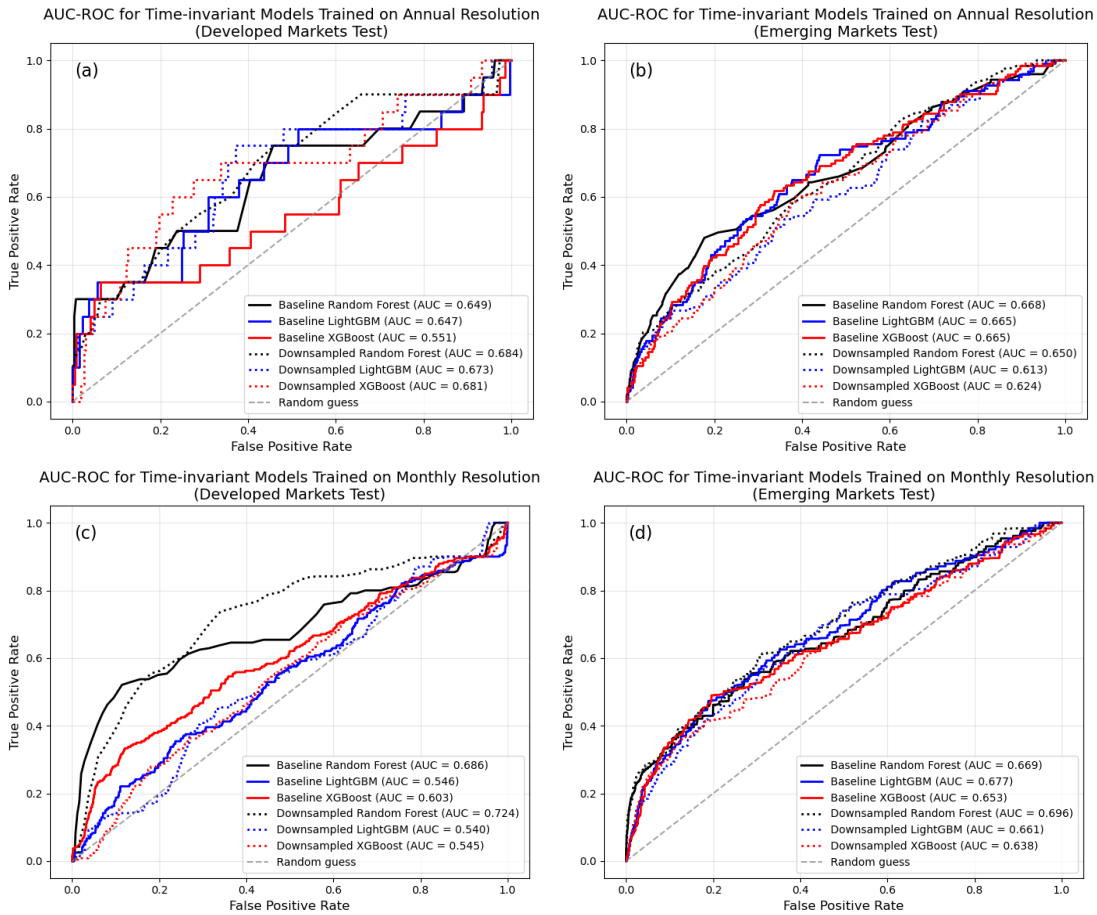

Figure 3: AUC-ROC of time-invariant models trained and tested on (a) annual resolution data for developed markets, (b) annual resolution data for emerging markets, (c) monthly resolution data for developed markets, (d) monthly resolution data for emerging markets.

The models exhibited a tendency to overpredict crisis labels for countries present in the OECD database which included all 17 developed markets and select emerging markets and made almost no positive predictions for other countries. This suggests that the higher AUC-ROC at monthly resolution is skewed by the high recall achieved for South Africa, Brazil, Chile, Colombia, Costa Rica, Mexico, China, India, Indonesia, and Turkey, which are the only emerging markets to have these monthly features. Overprediction was exaggerated for models trained on downsampled data due to the models overfitting to a small variety of negative crisis labels.

4.1.2 Dimensionality Reduction with Machine Learning Classifiers

To combat overfitting to a subset of markets with monthly data, methods of linear and non-linear dimensionality reduction were explored. This included Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding, Uniform Manifold Approximation and Projection, and Variational Autoencoders (VAEs). Figure 4 summarizes the effect of some dimensionality reduction methods on the performance of the classifier.

In the first case, PCA was applied to the training data before fitting the classifier. A second approach was to fit two separate PCA transformations, one for each class, and concatenate the reduced dimension data before fitting the classifier. In both instances, AUC-ROC was not better than using the original features; however, the models were able to make positive predictions for non-OECD countries.

A third approach was to train a VAE and perform classification using the encoded representation. This increased AUC-ROC for developed markets significantly but had no impact on emerging markets. This suggests that non-linear patterns have high predictive value and that the encoder learned to fit better to developed markets, likely because they exhibited greater variance at the monthly resolution than the emerging markets which were forward filled. In all the time-invariant cases, AUC-ROC improved when the model was trained on both developed and emerging markets compared to training on only developed markets.

4.2 Time-varying Models

Time-varying models can be trained on sequences of monthly data such that the crisis prediction is informed by the economic state of a country in relation to its recent history. This can be useful for a few reasons:

1. The model has the means to understand overarching trends and outlier periods in country-specific context.
2. The crisis label is positive for a continuous period of two years before the actual crisis; receiving a sequence of data can allow the model to better represent the positive period instead of overfitting to disparate monthly signals.
3. The crises themselves occur at low frequencies and can last a few years; receiving a sequence of data that includes in-crisis periods provides the model with information on recovery and can pre-empt the prediction of another crisis too soon after. For this reason, the time-varying models were trained on sequences that included in-crisis periods.

4.2.1 Dimensionality Reduction with Machine Learning Classifiers

One approach to time-varying modeling is to architect a dimensionality reduction method that can receive a sequence and appropriately embed it into a reduced dimension feature space for a machine learning classifer.

Figure 4 puts the AUC-ROC performance of a Temporal VAE embedding and Random Forest classification combination in context with other methods explored in the time-invariant case. Despite the low AUC-ROC, the model learns to make short, consecutive sequences of positive predictions instead of long periods of positive predictions. The limited overlap between the true labels and predictions shows that the model is still struggling to learn useful crisis precursors for emerging markets.

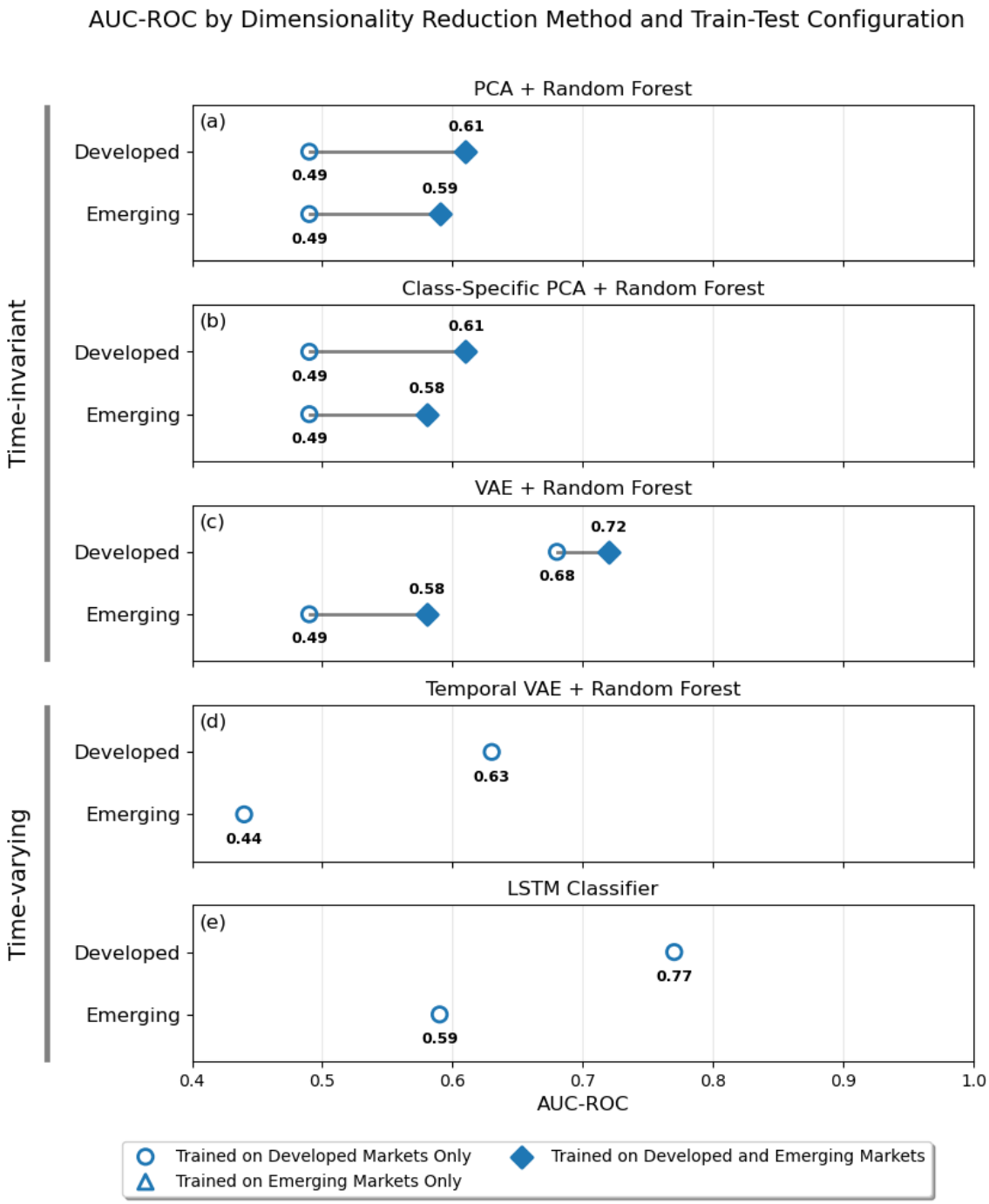

Figure 4: AUC-ROC of different dimensionality reduction and model configurations including (a) PCA + Random Forest, (b) Class-Specific PCA + Random Forest, (c) VAE + Random Forest, (d) Temporal VAE + Random Forest, (e) LSTM with Classifier head.

4.2.2 Deep Learning Classifiers

Tuning a Temporal VAE to improve classification performance was difficult as a high dimensional latent space was required to better represent the information embedded in the sequence of data yet the goal was to reduce the dimensions to prevent overfitting the classifier, as in the time-invariant case.

An alternative approach is to construct a deep learning model wherein classification forms the final layer. This can be trained end-to-end to learn the most relevant features from the sequence through attention mechanisms and residual connections and tuned using focal loss to draw attention to the difficult-to-classify sequences. A model trained on only developed markets and tuned using precision-based focal loss, defined in Equation 1 with $\alpha = 0.1$, $\gamma = 2.0$ and $\lambda = 2.25$, attained an AUC-ROC of 0.77 on developed markets and 0.59 on emerging markets in the test period (Figure 4).

$$\mathcal{L}_{\text{Precision Focal}}(p, y) = \underbrace{\alpha_t (1 - p_t)^{\gamma} \left(-\log p_t\right)}_{\text{focal loss}} + \underbrace{\lambda\, \sigma(z)\, (1 - y)}_{\text{false positive penalty}} \tag{1}$$

where $\alpha_t = \alpha\, y + (1 - \alpha)\,(1 - y)$, $p_t = y\, \sigma(z) + (1 - y)\,(1 - \sigma(z))$, and $\sigma(z) = \frac{1}{1+e^{-z}}$

The best performing architecture used multiple LSTM cells with attention and pooling layers before the classification head. Bidirectional LSTM captured long-term dependencies and temporal patterns from past and future context. Attention-weighted feature importance allowed the classifier to focus on the most relevant time periods, making the model more resilient to the choice of sequence length compared to previous time-varying experiments. Multiple pooling strategies including maximum, average, and last timestep provided the classifier with information on current and overall economic health and residual connections aided in end-to-end information flow from data input to classification.

This model offered the best out-of-the-box performance on the emerging markets dataset. This supports a claim that data on developed markets can provide some useful crisis precursors for emerging markets, referred to as "monsoonal effects" [23], as major economic shifts in developed markets can trigger crises in emerging markets. There are also "spillover effects" [23], resulting from interdependence among emerging markets. Transfer learning showed limited ability in capturing these and often undermined the performance of the classifier. Figure 5 shows the highest AUC-ROC achieved with different proportions of trainable layers. The best model, entirely re-trained on emerging markets with unique values for α, γ, and λ, was only slightly more performant.

Layer Group	% Trainable Layers 10%	25%	50%	100%
Input layers				✓
LSTM layers (l_0)				✓
LSTM layers (l_1)				✓
LSTM layers (l_2)			✓	✓
LSTM batch normalization			✓	✓
Attention mechanism			✓	✓
Attention normalization		✓	✓	✓
Dense blocks (block 0)		✓	✓	✓
Dense blocks (block 1)	✓	✓	✓	✓
Residual connections				✓
Classifier	✓	✓	✓	✓

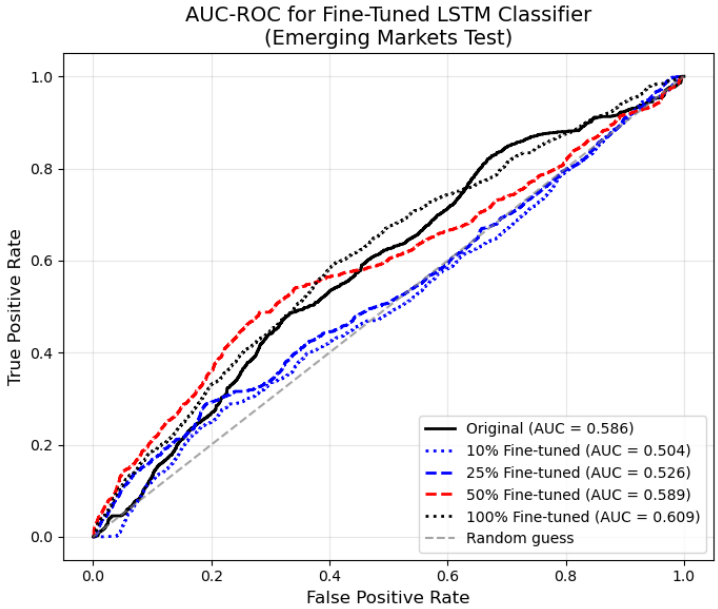

Figure 5: *Top*: Description of the trainable layers in each experiment. *Bottom*: Best AUC-ROC achieved by transfer learning with different proportions of the LSTM Classifier layers on emerging markets.

5 Recommendations

Here, we re-iterate the data availability and quality gap that exists between developed and emerging markets and discuss how this undermines the performance of models on emerging markets. We also suggest ways to bridge this gap using domain knowledge and data-driven methods.

5.1 A high-dimensional feature space risks overfitting

This problem suffers from a strong degree of class imbalance and high dimensionality (Table 2). This resulted in models overfitting to small signatures of noise and being unable to generalize well to unseen crises. Downsampling the majority class to address class imbalance meant that models were trained on less diverse samples of normal economic periods. Even though AUC-ROC was often greater (see Figure 3), the models overpredicted crises in the test set with low precision and high recall.

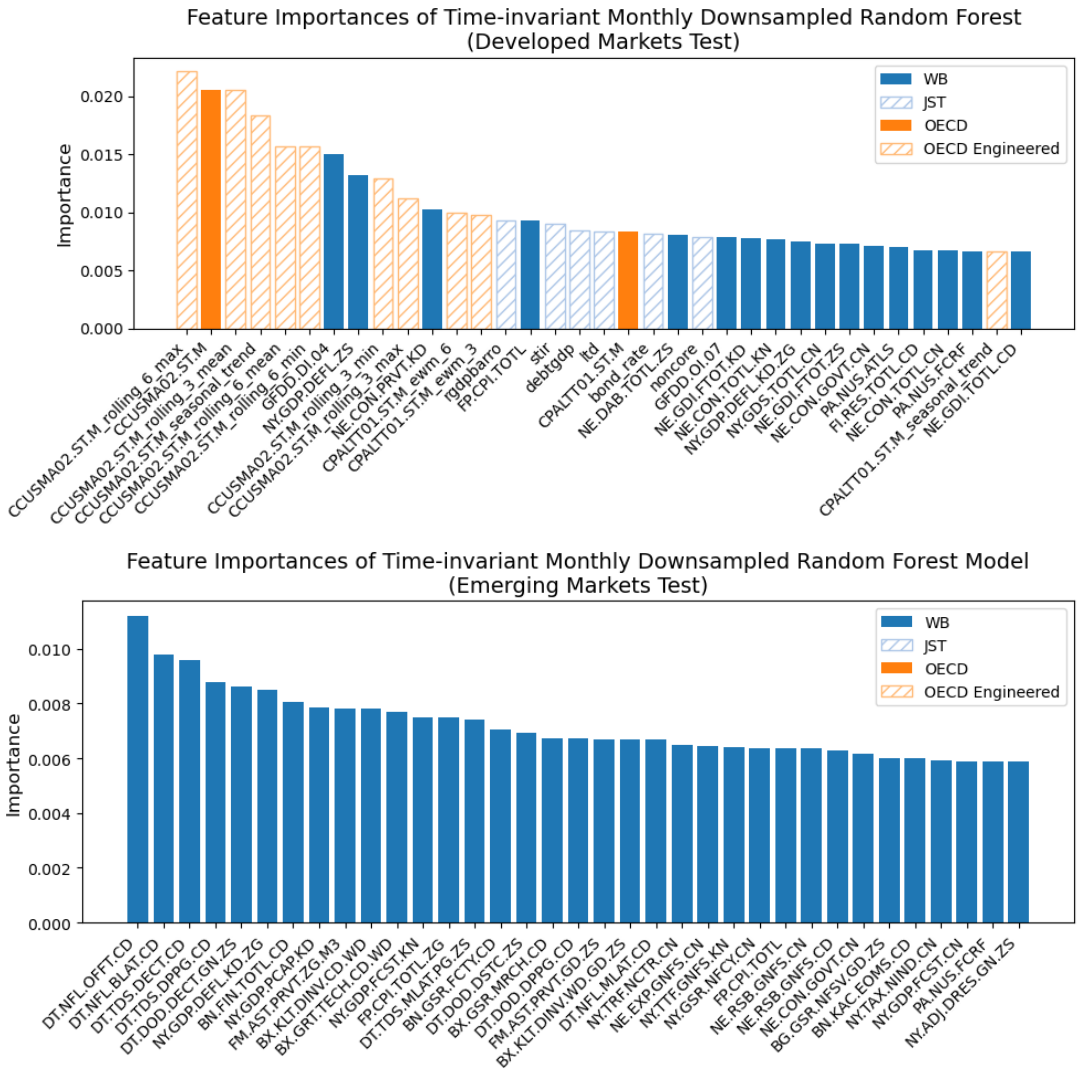

Figure 6: Feature importances of time-invariant machine learning classifiers with the highest AUC-ROC for each market. *Top*: Random Forest model trained on downsampled monthly data for developed markets. *Bottom*: Random Forest model trained on downsampled monthly data for emerging markets.

Performance on emerging markets was consistently lower than on developed markets. Figure 6 shows that this is because the most important features were the monthly resolution OECD variables, the features engineered from them (OECD Engineered), and high quality annual JST variables, which were either underrepresented in or wholly unreported for emerging markets. Training models specific

to emerging markets was necessary to achieve any reasonable performance. Even so, models overfit to OECD countries which had the highest quality data.

Recommendation 1: Improve quality of predictors for emerging markets. Feature importances show that high quality OECD-like features can improve model performance. Emerging markets that are missing these dimensions can borrow them from the OECD countries that belong to the same geographical region, have similar economic structures, share historical crisis years, or are major trade partners. For instance, `CCUSMA02.ST.M` is the monthly average of the exchange rate per USD and shows high predictive value for crises in developed markets (Figure 6). The last five financial crises in Ghana occurred within a few years of crises in South Africa, the only OECD country in the region. South Africa is also a major export and import partner [1]. Monthly `CCUSMA02.ST.M` features can be created for Ghana by using the annual relationship in exchange rates between Ghana and South Africa and interpolating to a monthly frequency.

5.2 Pre-crisis periods are not easily separable from other economic states

Figure 7 shows that PCA, when applied individually to the positive and negative labels, requires fewer principal components to explain the variance in the positive label. This is because the positive label represents only a small fraction of all time, the two years pre-crisis, while the negative class contains a wide range of economic states including fast or slow economic growth, plateaus, minor (non-crisis) economic downturns, and even recovery. The first components of both classes are also highly correlated as they represent similar underlying features.

PCA is good at separating countries where data is sufficiently country-specific (e.g. United States and Japan). However, it fails to separate the positive from the negative labels as both appear to occupy the same space and belong to the same clusters. Non-linear, deep learning methods such as VAEs also struggle with this.

In the time-invariant case, dimensionality reduction was applied post removal of the in-crisis period to narrow the economic states present in the negative class and encourage separation. Additional features such as rolling statistics, exceedances, momentum, volatility were engineered to encapsulate temporal context. These features were deemed important by the downstream classification models (Figure 6) but they did not sufficiently improve separation of the classes in the feature space. A more refined approach could employ a country-specific methodology to remove in-crisis periods and determine more useful periodicities of temporal features (e.g. informed by the country's political election cycles, fiscal and monetary policy changes).

In the time-varying case, it was hypothesized that retaining the full temporal context, including in-crisis periods, could allow a model to archetype economic trends with the recurrent unit handling sequential dependencies. This was tested as an alternative dimensionality reduction method and as an end-to-end embedding and classification approach but only in the latter case did it substantially improve performance. The embedding space of the best performing LSTM Classifier model shows more obvious separation of the positive label cluster from the negative label, highlighting that performance is related to the quality of the feature space.

Recommendation 2: Tune feature space to maximize separation of positive and negative crisis labels. We can encourage improved separation of crisis labels to make the embedding feature space more useful to the downstream classification task. Silhouette score or density-based cluster validation can be used to assess the quality of positive and negative label clusters and the separation between the two. Tuning with these metrics can enhance the quality of the feature space and potentially augment the learning process. This is particularly useful in encoder architectures where the trade-off is between reconstruction (i.e. data representation) and KL Divergence (i.e. generalizability) but reconstruction

[1] Ghana Trade Summary 2022. World Integrated Trade Solution. `https://wits.worldbank.org/CountrySnapshot/en/GHA`

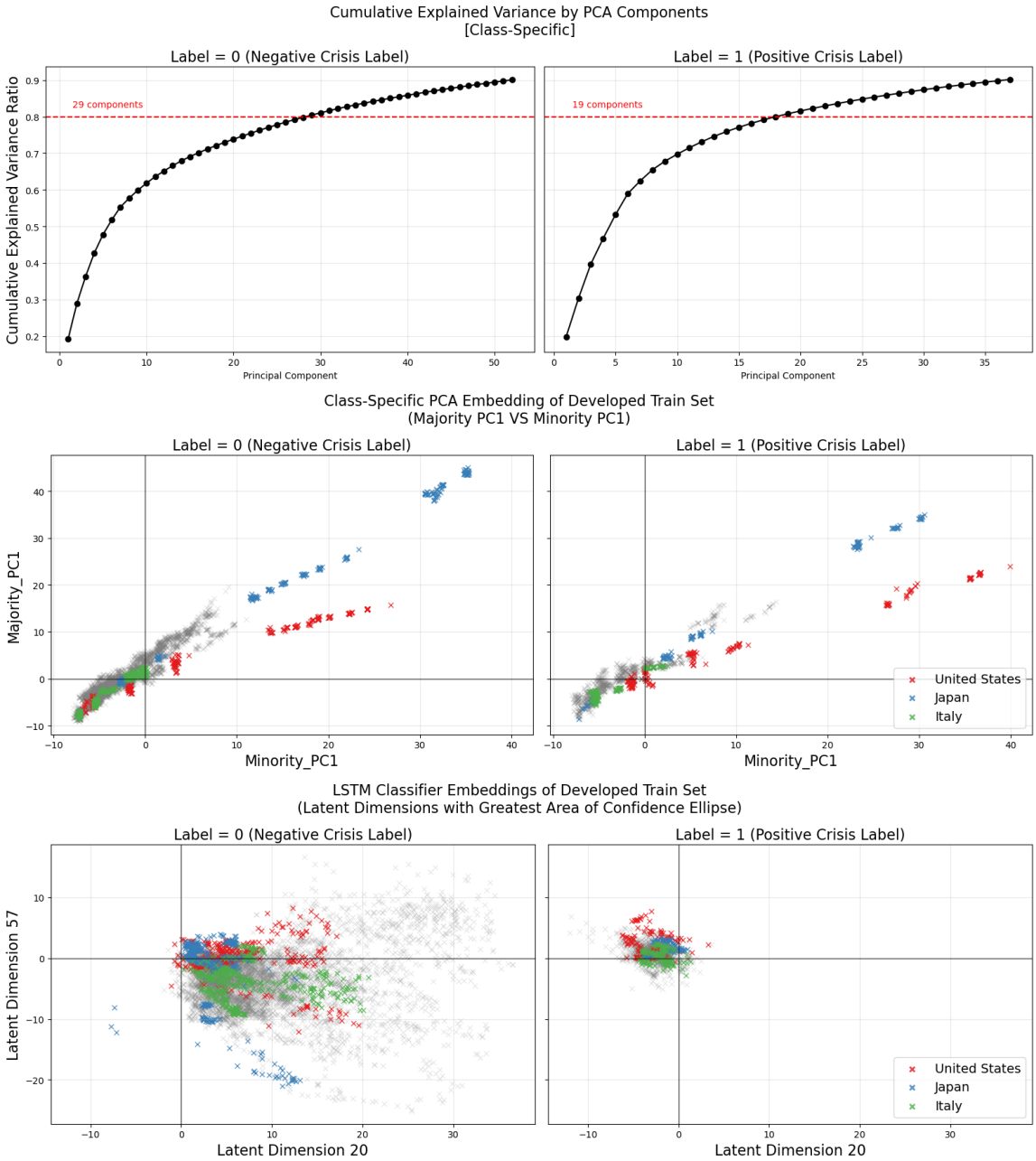

Figure 7: 2D projection of developed markets in the feature spaces created by Class-Specific PCA and the LSTM Classifier. Training data with negative crisis label is visualized on the left and positive crisis label on the right. *Top*: Cumulative explained variance of PCA components for positive and negative crisis labels separately after dropping in-crisis years.

need not be a primary goal. In fact, the best performing PCA and Random Forest combination was one trained on components that explained only 60% of the variance in the data.

5.3 Class imbalance inhibits the learning of meaningful crisis precursors

After dimensionality reduction and feature engineering, the models tested still struggled with class imbalance. The positive crisis labels are sparse in time. This makes it difficult to learn useful crisis precursors, particularly for emerging markets where data is limited and the monthly resolution was created by forward filling annual data.

In the time-varying deep learning case, the LSTM Classifier was tuned on focal loss. The best performing model was one trained on precision-based focal loss (Equation 1) with low α, so that the model was selective about when it predicted a positive crisis label, and high γ, which emphasized learning on the difficult-to-classify examples.

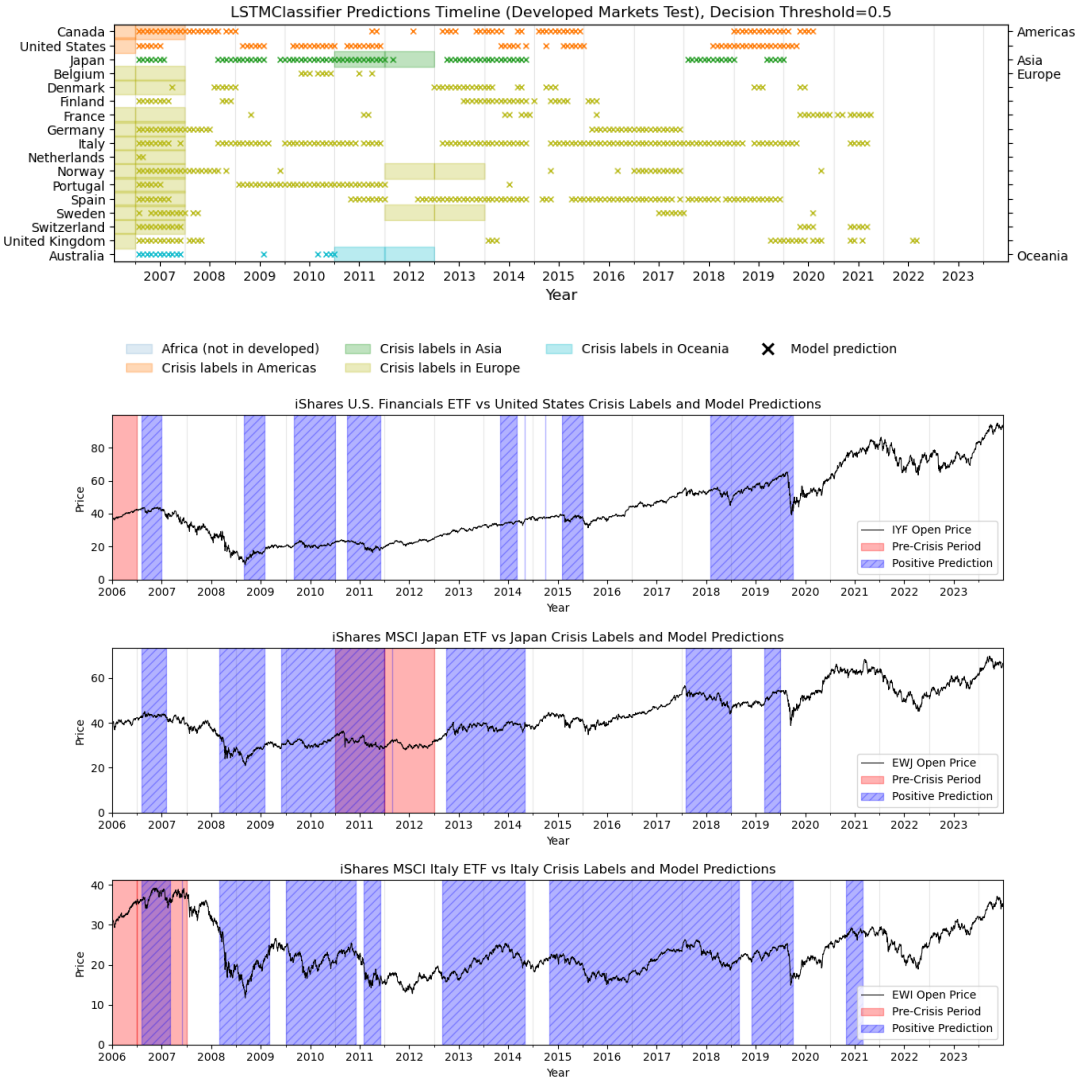

Figure 8: LSTM Classifier performance on developed markets in the test period, 2007-01-01 to 2023-12-31. Regions shaded in red denote the positive crisis label to be predicted. Regions in blue denote the model's positive predictions. *Top*: All developed markets. *2nd Row*: United States, overlaid on IYF, iShares U.S. Financials ETF. *3rd Row*: Japan, overlaid on EWJ, iShares MSCI Japan ETF. *Bottom*: Italy overlaid on EWI, iShares MSCI Italy ETF.

Figure 8 shows the performance of the model on predicting crises in developed markets during the test

period. The model predicts crises for several countries in 2007 but offers several additional positive predictions in subsequent years that are not flagged by the ground truth sources. For Canada, United States, Italy, and Portugal, the model predicts a positive label before the COVID-19 recession. This was not flagged as a crisis by the databases consulted. Positive predictions were also made for 2008 to 2009, an in-crisis and recovery period.

The positive predictions for the U.S. in 2010 to 2011 could be related to the high unemployment rate, which peaked at 10% in October 2009 and was still above 8.5% at the end of 2011 [2]. This was a slower recovery compared to after COVID, when unemployment fell from 14.8% to the pre-COVID 3.9% in less than two years. This could be why COVID recovery is not highlighted by the model.

Japan and Italy have more sustained and concurrent positive predictions from 2007 to 2011, which may reflect the prolonged aftermath of the Global Financial Crisis in these economies. Interestingly, the model predicts a crisis for Japan in 2013-2014, during which the country was practicing "Abenomics" [17]. This was a favorable period during which foreign investors purchased Japanese stocks and sold Japanese yen. The model, which instead observed local economic features, likely flagged the volatility of exchange rates as a crisis precursor. Italy experienced a soverign debt crisis shortly after the Global Financial Crisis. GDP only exceeded pre-2007 levels in 2024 [12].

In sum, there are some defensible signals of crisis precursors that the model has learned to identify but several false positives cannot be explained. In general, the lingering false positives are common for economies with slower recoveries, shaped by external conditions, and with pronounced signals of distress. This includes many emerging markets.

Recommendation 3: Enrich the minority class by predicting almost-crises. Instead of focusing on periods classified as major financial crises by economists, additional labels can be created for periods of instability or heightened risk to enrich the positive labels. These can originate from financial literature or from trading price signals of market-specific investments. For example, the Italian soverign bond spread experienced high volatility in 2012, before Greece's debt crisis which had spillover effects [13]. This is heightened risk that models can be encouraged to identify.

[2] U.S. Bureau of Labor Statistics. `https://www.bls.gov/charts/employment-situation/civilian-unemployment-rate.htm`

6 Conclusion

This study demonstrates the challenges of predicting financial crises in emerging markets using economic data from the World Bank and OECD. Data is often only available at annual resolution for these markets, which limits the ability of machine and deep learning models in capturing the granular economic dynamics that exhibit high predictive value for crisis prediction in developed markets. Models trained on developed markets often predict crisis concurrence and lingering false positives for emerging markets. This is systematic bias with high risk of misuse off-the-shelf for fragile economies.

Limited availability of quality macroeconomic data is a challenge that plagues domain-specific modelling for emerging markets. More recently, there is a push to leverage textual data from news articles, social media, policy announcements, and other sources to enrich the data space. Studies show that some macroeconomic movements are more linked with sentiment than with traditional indicators [13] and that deep learning can extract some predictive economic features from textual data [8]. Incorporating textual data can push the boundaries of context-awareness and multi-modality for quality model development, particularly for emerging markets which are shaped by external conditions and speculation.

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